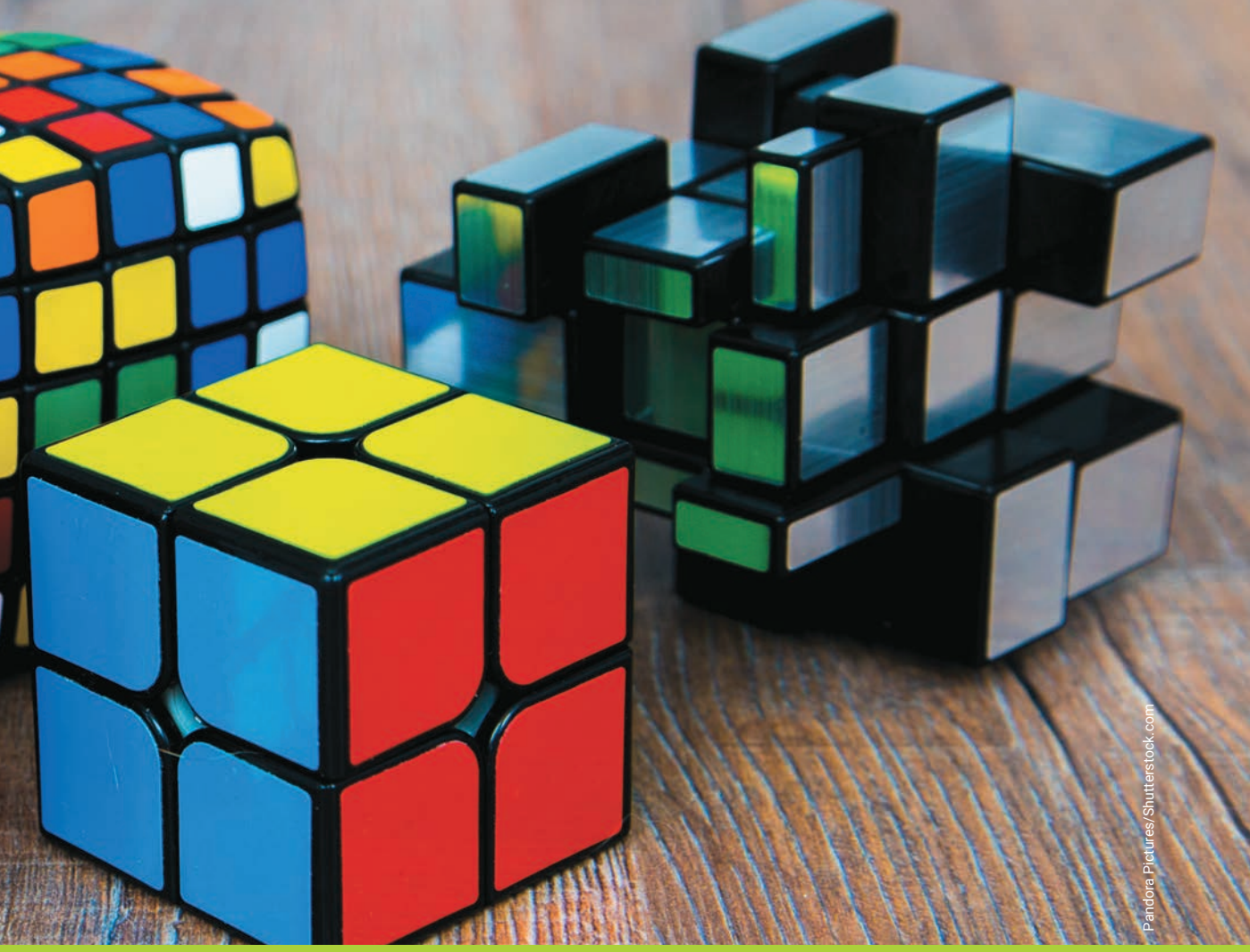


4

NUMBER AND ALGEBRA

ALGEBRA

Rubik's Cube is a puzzle that was invented by Hungarian Professor of Architecture Erno Rubik in 1974. In Hungary, it was originally called the Magic Cube.
The cube has $2^{10} \times 3^7 \times 8! \times 12! \approx 43\,252\,003\,274\,489\,856\,000$ different possible arrangements (8! or '8 factorial' means $8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1$). In 2023, US 'speedcuber' Max Park solved a Rubik's cube puzzle in 33 moves in 3.134 seconds!



Chapter outline

		Working mathematically				
4.01	The index laws	U	F		R	C
4.02	Adding and subtracting algebraic fractions	U	F			
4.03	Multiplying and dividing algebraic fractions	U	F			
4.04	Expanding and factorising expressions	U	F			
4.05	Expanding binomial products	U	F		R	C
4.06	Factorising quadratic expressions $x^2 + bx + c$*	U	F		R	

*PATH

Wordbank

binomial An algebraic expression that consists of 2 terms, for example, $4a + 9$, $3 - y$, $x^2 - 4x$

binomial product Binomials multiplied together, for example, $(x + 9)(3x - 4)$

expand To rewrite an expression such as $5(2k - 6)$ without grouping symbols; for example, $5(2k - 6)$ expands to $10k - 30$

factorise To rewrite an expression with grouping symbols, by taking out the highest common factor. Factorising is the opposite of expanding; for example, $9r^2 + 36r$ factorised is $9r(r + 4)$

index laws Rules for simplifying algebraic expressions involving powers of the same base; for example, $a^m \div a^n = a^{m-n}$

quadratic expression An algebraic expression in which the highest power of the variable is 2; for example, $2x^2 + 5x - 3$ or $x^2 + 2$

quadratic trinomial A quadratic expression that has 3 terms, for example, $2x^2 + 5x - 3$

Quiz
Wordbank 4

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (11):

4.01 Numbers and powers • Simplifying with the index laws • Negative indices • Index laws
4.02 Adding and subtracting algebraic fractions • Algebraic fractions
4.03 Algebraic fractions
4.04 Factorising expressions 2
4.05 Expanding binomial products 1 • Expanding binomial products 2
4.06 Factorising quadratic expressions 1

Quizzes (5):

- Wordbank 4
- SkillCheck 4
- Mental skills 4
- Language of maths 4
- Test yourself 4

Skillsheets (3):

4.04 Algebra using diagrams • HCF by factor trees • Factorising using diagrams

Worksheets (9):

4.01 Index laws review
4.03 Algebraic fractions
4.04 Expanding and factorising • Algebra 4
4.05 Mixed expansions • Algebra 2 • Binomial products • Area diagrams
Mind map Algebra

Puzzles (10):

4.01 Indices puzzle • Indices squaresaw
4.02 Algebraic fractions
4.03 Algebraic fractions puzzle • Upside-down fractions
4.04 Factorising puzzle • The distributive law
4.05 Expanding brackets • Trinominoes
Language of maths Algebra crossword

Technology (1):

4.05 Expanding binomials

Spreadsheet (1):

4.05 Expanding binomials

Presentation (1):

4.01 Index laws

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

In this chapter you will:

- apply index laws to numerical expressions with integer indices
- (PATH) apply index laws to algebraic expressions with integer indices
- interpret and use zero and negative indices
- add, subtract, multiply and divide simple algebraic fractions
- (PATH) add, subtract, multiply and divide algebraic fractions with pronumerals in the denominator
- expand and factorise algebraic expressions
- (PATH) factorise algebraic expressions involving terms with indices
- expand binomial products
- (PATH) expand special binomial products
- (PATH) factorise quadratic expressions of the form $x^2 + bx + c$

SkillCheck ANSWERS ON P. 585

Quiz
SkillCheck 4

4

1 Simplify each expression.

a $g^4 \times g^5$ **b** $r^8 \div r^2$ **c** $(d^5)^3$

d $(-k)^2$ **e** $h \times h^9$ **f** $m^5 \div m$

g a^1 **h** a^0 **i** $3e^2 \times 2e^4$

j $18n^6 \div 6n^2$ **k** $(10w^3)^3$ **l** $25q^0$

m $(vw)^5$ **n** $\left(\frac{c}{p}\right)^3$ **o** y^{-1}

p k^{-2} **q** $\frac{35ad}{20a^2}$ **r** $4y^2 \times \frac{3}{8}y^3$

s $3g^{-2} \times 8g^4$ **t** $\left(\frac{2b}{3h}\right)^2$

2 Evaluate each expression.

a $\frac{a}{5} + \frac{3a}{4}$ **b** $\frac{7p}{2} - \frac{10p}{3}$ **c** $\frac{8}{w} \times \frac{5w}{24}$ **d** $\frac{x}{14} \div \frac{y}{2}$

3 Expand each expression.

a $6m(3m + 11)$ **b** $-5(3g - 8)$ **c** $3w(8y - 4w)$

4 Factorise each expression.

a $4x + 24$ **b** $20 - 15a$ **c** $q^2 + q$

d $18a^2 - 12a$ **e** $-2y - 30$ **f** $-18w + 24$

5 Find 2 numbers whose:

a product is 18 and sum is 9

b product is 8 and sum is -6

c product is -20 and sum is -1

d product is -16 and sum is 6.

4.01 The index laws

Index is another name for **power**. The plural of index is **indices** (pronounced 'in-de-sees'). The following rules are called **index laws**.

Videos
Numbers and powers
Simplifying with the index laws
Negative indices
Index laws

Worksheet
Index laws review

Presentation
Index laws

PATH

The index laws

When **multiplying terms with the same base**, add the powers

$$a^m \times a^n = a^{m+n}$$

When **dividing terms with the same base**, subtract the powers

$$a^m \div a^n = \frac{a^m}{a^n} = a^{m-n}$$

When **raising a term with a power to another power**, multiply the powers

$$(a^m)^n = a^{m \times n}$$

When **raising a product of terms to a power**, raise each term to that power

$$(ab)^n = a^n b^n$$

When **raising a quotient of terms to a power**, raise each term to that power

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

Any number **raised to the power of zero** is equal to 1

$$a^0 = 1$$

A number **raised to a negative power** gives a fraction (with a numerator of 1)

$$a^{-n} = \frac{1}{a^n}$$

A number **raised to a power of −1** gives its reciprocal

$$a^{-1} = \frac{1}{a}$$

$$\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}$$

Puzzles
Indices puzzle
Indices squaresaw

Example 1

Simplify each expression.

a $5p^2 q^3 \times 4p^3 q^4$ **b** $\frac{24e^6n^{12}}{8en^4}$ **c** $(3t^5)^2$

d $(4d^4 q^2 r)^3$ **e** $\left(\frac{2c^2}{d}\right)^4$ **f** $31x^0 - (31x)^0$

Solution

a $5p^2 q^3 \times 4p^3 q^4 = 5 \times 4 \times p^2 \times p^3 \times q^3 \times q^4$
$= 20p^{2+3} q^{3+4}$
$= 20p^5 q^7$

b $\frac{24e^6n^{12}}{8en^4} = \frac{{}^{3}\cancel{24}e^6n^{12}}{{}_{1}\cancel{8}en^4}$
$= 3e^{6-1} n^{12-4}$
$= 3e^5 n^8$

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c $(3t^5)^2 = 3t^5 \times 3t^5$ or $(3t^5)^2 = 3^2 \times t^{5\times 2}$
$= 9t^{5+5}$ $= 9t^{10}$
$= 9t^{10}$

d $(4d^4 q^2 r)^3 = 4^3 d^{4\times 3} q^{2\times 3} r^3$
$= 64d^{12} q^6 r^3$

e $\left(\frac{2c^2}{d}\right)^4 = \frac{(2c^2)^4}{d^4}$
$= \frac{2^4 c^{2\times 4}}{d^4}$
$= \frac{16c^8}{d^4}$

f $31x^0 - (31x)^0 = 31 \times 1 - 1$
$= 30$

4.01

PATH

Example 2

Simplify each expression using a positive index.

a m^{-3} **b** $9x^{-2}$ **c** $(4q)^{-3}$ **d** $\left(\frac{2}{3}\right)^{-1}$

Solution

a $m^{-3} = \frac{1}{m^3}$

b $9x^{-2} = 9 \times x^{-2}$
$= 9 \times \frac{1}{x^2}$
$= \frac{9}{x^2}$

c $(4q)^{-3} = \frac{1}{(4q)^3}$
$= \frac{1}{64q^3}$

d $\left(\frac{2}{3}\right)^{-1} = \left(\frac{3}{2}\right)^1$
$= \frac{3}{2}$

Negative powers of quotients

Consider $\left(\frac{4}{5}\right)^{-2} = \frac{1}{\left(\frac{4}{5}\right)^2} = \frac{1}{\frac{16}{25}}$
$= 1 \div \frac{16}{25}$
$= 1 \times \frac{25}{16}$
$= \frac{25}{16}$
$= \frac{5^2}{4^2}$
$= \left(\frac{5}{4}\right)^2$

Negative powers of quotients

A number **raised to a power of $-n$** gives its reciprocal raised to the power of n.

$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n = \frac{b^n}{a^n}$$

PATH

Example 3

Simplify each expression.

a $\left(\frac{4}{3}\right)^{-3}$ **b** $\left(2\frac{1}{2}\right)^{-2}$ **c** $\left(\frac{3a}{b^3}\right)^{-4}$

Solution

a $\left(\frac{4}{3}\right)^{-3} = \left(\frac{3}{4}\right)^{3}$

$= \frac{27}{64}$

b $\left(2\frac{1}{2}\right)^{-2} = \left(\frac{5}{2}\right)^{-2}$

$= \left(\frac{2}{5}\right)^{2}$

$= \frac{4}{25}$

c $\left(\frac{3a}{b^4}\right)^{-2} = \left(\frac{b^4}{3a}\right)^{2}$

$= \frac{b^8}{9a^2}$

EXERCISE 4.01 ANSWERS ON P. 586

The index laws U F R C

EXAMPLE 1

1 Simplify each expression. R C

a $5k^4 \times 3k^7$ **b** $30y^{10} \div 3y^7$ **c** $12p^6 \times \frac{3}{4}p^5$

d $(w^3)^5$ **e** $(2n^3)^5$ **f** $\frac{24h^5}{8h}$

g $3a^2d \times 4a^5d^4$ **h** $(3q^3)^2$ **i** $32w^5y^8 \div 8w^3y^4$

j $(-2c^2)^7$ **k** $\frac{28a^2c^5d^7}{21c^4d^4}$ **l** $(5y^6)^3$

m $4h^3k^2 \times 6hk^3w^5$ **n** $(2d^3g^2)^5$ **o** $\frac{8m^5p^{11}q}{12m^3p^7q}$

2 Simplify each expression. R C

a $(l^3m^5)^6$ **b** $\left(\frac{n}{2}\right)^3$ **c** $7x^0$ **d** $\left(\frac{w^2}{k^3}\right)^5$

e $\left(\frac{2}{3}\right)^0$ **f** $(-8ky^5)^2$ **g** $(16a)^0 - 16a^0$ **h** $\left(\frac{2b}{3d}\right)^4$

i $(5xy^2)^0$ **j** $(-5dy^2)^4$ **k** $\left(-\frac{3k^4}{10}\right)^3$ **l** $-9(a^2b^3)^0$

m $(2p^2q^3r^4)^4$ **n** $(4w^0)^3$ **o** $(-3g^6k)^2$ **p** $3a^0 + (3ab)^0$

3 Evaluate each expression. R C

a 4^0 **b** $(-8)^0$ **c** 7×2^0 **d** $(5 \times 3)^0$

e $(-2)^3$ **f** $(-3)^2$ **g** $(5^2)^2$ **h** $4^0 + 7^0$

i $2^4 \times 2^3$ **j** $(10^2)^0$ **k** $(2^3)^0 - (2^0)^3$ **l** $4^5 \div 4^2$

m $6^2 \div 6^5$ **n** $9^5 \div 9^5$ **o** $(8 \times 3)^0 - 8 \times 3^0$ **p** $5^2 \div 5^0$

q $\left(\frac{2}{5}\right)^0 + \left(\frac{5}{2}\right)^0$ **r** $3^{-2} \div 3^2$ **s** $\left(\frac{3}{4}\right)^0$ **t** $12^{-2} \times 12^2$

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4 Find the value of $(3x)^0 + 3 \times 2^0$. Select the correct answer **A**, **B**, **C** or **D**.

A 6 **B** 1 **C** 7 **D** 4

5 Express each term as a fraction. R C

a 5^{-2} **b** 2^{-5} **c** 20^{-1} **d** 10^{-3} **e** 3^{-4}

6 Simplify $2c^{-3}$. Select **A**, **B**, **C** or **D**.

A $-6c$ **B** $\frac{1}{2c^3}$ **C** $\frac{2}{c^3}$ **D** $\frac{1}{8c^3}$

7 Simplify each expression using positive indices. R C

a 8^{-7} **b** 3^{-5} **c** y^{-1} **d** x^{-3}

e $(5b)^{-2}$ **f** $8h^{-3}$ **g** $(ab)^{-1}$ **h** $-pq^{-1}$

i $11w^{-3}$ **j** $(6x)^{-3}$ **k** $a^3 b^{-5}$ **l** mw^{-3}

m $8u^{-3} v^{-4}$ **n** $-2r^6 y^{-5}$ **o** $10e^{-1} f^3$ **p** $\frac{1}{2} g^4 h^{-3}$

q $\frac{3}{4} d^7 n^{-2}$ **r** $(-4c)^{-2}$ **s** $5x^2 y^{-1} w^{-2}$ **t** $2(mp)^{-1}$

8 Simplify each expression. R C

a $10r^{-6}$ **b** $\left(\frac{5}{2}\right)^{-1}$ **c** $\left(\frac{1}{3}\right)^{-1}$ **d** $\left(\frac{1}{x}\right)^{-1}$

e $\left(\frac{k}{3}\right)^{-1}$ **f** $5a^{-2}$ **g** $(6w)^{-2}$ **h** $\left(-\frac{m}{2}\right)^{-1}$

i $\left(\frac{5}{3g^3}\right)^{-1}$ **j** $\left(\frac{2r}{3h}\right)^{-1}$ **k** $m^3 n^2 p^{-2}$ **l** $\left(\frac{5b}{4a^2}\right)^{-1}$

9 Simplify each expression, writing your answer with positive indices. R C

a $(2x^5 y^2)^2 \times 5x^{-4} y^5$ **b** $(3m^8 n^{-2} \times m^{-2} n^3)^4$

c $\left(\frac{6k^3 w^5}{9k^4 w^3}\right)^2$ **d** $(5g^6 y^4)^2 \div 10g^8 y^{-4}$

e $\left(\frac{2a^4 x^4}{3a^5 x^2}\right)^{-3}$ **f** $-6a^2 d^3 \times (-2a^3 d^4)^3$

g $3q^{-5} r^3 \div (6qr^2)^2$ **h** $(4h^3 k^2)^3 \times (hk^5)^2$

i $[24b^5 q^6 \div (-2bq^2)^3]^2$

10 Simplify each expression. R C

a $\left(\frac{3}{4}\right)^{-2}$ **b** $\left(\frac{2}{3}\right)^{-3}$ **c** $\left(-\frac{1}{10}\right)^{-6}$ **d** $\left(\frac{5}{2}\right)^{-3}$

e $\left(-\frac{4}{3}\right)^{-5}$ **f** $\left(\frac{5}{4}\right)^{-4}$ **g** $\left(2\frac{1}{4}\right)^{-2}$ **h** $\left(1\frac{2}{5}\right)^{-3}$

i $\left(\frac{k}{3}\right)^{-2}$ **j** $\left(-\frac{3}{x}\right)^{-3}$ **k** $\left(\frac{a^2}{5}\right)^{-4}$ **l** $\left(\frac{4}{3g^3}\right)^{-2}$

4.01

PATH

EXAMPLE 2

EXAMPLE 3

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4.02 Adding and subtracting algebraic fractions

Videos
Adding and subtracting algebraic fractions

Algebraic fractions

Puzzle
Algebraic fractions

Adding and subtracting algebraic fractions

To add or subtract fractions, convert them (if needed) so that they will have the same denominator, then simply add or subtract the numerators.

Example 4

Simplify each expression.

a $\frac{a}{2}+\frac{a}{3}$ **b** $\frac{2x}{5}-\frac{x}{3}$ **c** $\frac{5}{6}+\frac{7b}{15}$

Solution

a $\frac{a}{2}+\frac{a}{3}=\frac{3\times a}{3\times 2}+\frac{2\times a}{2\times 3}$

$=\frac{3a}{6}+\frac{2a}{6}$

$=\frac{5a}{6}$

Common denominator $=2\times 3=6$

b $\frac{2x}{5}-\frac{x}{3}=\frac{3\times 2x}{3\times 5}-\frac{5\times x}{5\times 3}$

$=\frac{6x}{15}-\frac{5x}{15}$

$=\frac{x}{15}$

c $\frac{5}{6}+\frac{7b}{15}=\frac{25}{30}+\frac{14b}{30}$

$=\frac{25+14b}{30}$

The lowest common denominator is 30

EXERCISE 4.02 ANSWERS ON P. 586

Adding and subtracting algebraic fractions UFRC

EXAMPLE 4

1 Simplify each expression.

a $\frac{2k}{7}+\frac{3k}{7}$ **b** $\frac{8x}{11}-\frac{5x}{11}$ **c** $\frac{13m}{15}-\frac{7m}{15}$ **d** $\frac{9g}{5}+\frac{4g}{5}$

e $\frac{3w}{4}-\frac{2w}{5}$ **f** $\frac{a}{3}+\frac{3a}{7}$ **g** $\frac{4c}{5}+\frac{7c}{12}$ **h** $\frac{5u}{2}-\frac{u}{6}$

PATH

i $\frac{3}{4k}-\frac{1}{4k}$ **j** $\frac{9}{7x}+\frac{5}{7x}$ **k** $\frac{13}{5h}-\frac{3}{5h}$ **l** $\frac{5}{m}+\frac{7}{2m}$

m $\frac{5}{3a}-\frac{1}{a}$ **n** $\frac{7}{2h}+\frac{4}{3h}$ **o** $\frac{11}{8k}+\frac{4}{5k}$ **p** $\frac{16}{7p}-\frac{5}{3p}$

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2 Simplify each expression.

a $\frac{m}{3}+\frac{n}{5}$ **b** $\frac{k}{7}-\frac{w}{6}$ **c** $\frac{x}{4}-\frac{p}{9}$ **d** $\frac{5}{8}+\frac{h}{3}$

e $\frac{n}{10}-\frac{4}{7}$ **f** $\frac{5y}{2}-\frac{2c}{5}$ **g** $\frac{3q}{11}+\frac{2d}{3}$ **h** $\frac{3a}{8}+\frac{5e}{6}$

i $\frac{2b}{9}+\frac{b}{5}$ **j** $\frac{9y}{7}-\frac{3y}{4}$ **k** $\frac{5c}{3}-\frac{2c}{11}$ **l** $\frac{8g}{5}+\frac{2g}{15}$

m $\frac{4h}{3}+\frac{h}{8}$ **n** $\frac{10x}{7}+\frac{4x}{5}$ **o** $\frac{12}{5}-\frac{8k}{3}$ **p** $\frac{7m}{12}-\frac{2m}{15}$

Multiplying and dividing algebraic fractions

4.03

Multiplying and dividing algebraic fractions

- To multiply fractions, cancel any common **factors**, then multiply the numerators and denominators separately.
- To divide by a fraction $\frac{a}{b}$, multiply by its **reciprocal** $\frac{b}{a}$.

Worksheet Algebraic fractions

Puzzles Algebraic fractions puzzle

Upside-down fractions

Video Algebraic fractions

Example 5

Simplify each expression.

a $\frac{3x}{4}\times\frac{2x}{9}$ **b** $\frac{4}{k}\times\frac{3k}{16}$ **c** $\frac{2}{v}\div\frac{3}{w}$ **d** $\frac{xy}{5}\div\frac{3x}{25}$

Solution

a $\frac{3x}{4}\times\frac{2x}{9}=\frac{3x\times 2x}{4\times 9}$ or $\frac{3x}{4}\times\frac{2x}{9}=\frac{{}^{1}\cancel{3}x}{{}_{2}\cancel{4}}\times\frac{{}^{1}\cancel{2}x}{{}_{3}\cancel{9}}$

$=\frac{6x^2}{36}$ $=\frac{x^2}{6}$

$=\frac{x^2}{6}$

b $\frac{4}{k}\times\frac{3k}{16}=\frac{{}^{1}\cancel{4}}{\cancel{k}}\times\frac{3\cancel{k}}{16_{4}}$

$=\frac{3}{4}$

c $\frac{2}{v}\div\frac{3}{w}=\frac{2}{v}\times\frac{w}{3}$

$=\frac{2w}{3v}$

d $\frac{xy}{5}\div\frac{3x}{25}=\frac{\cancel{x}y}{\cancel{5}_{1}}\times\frac{\overset{5}{\cancel{25}}}{3\cancel{x}}$

$=\frac{5y}{3}$

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EXERCISE 4.03 ANSWERS ON P. 586

Multiplying and dividing algebraic fractions U F

EXAMPLE 5

1 Simplify each product.

a $\frac{m}{5}\times\frac{2}{7}$ **b** $\frac{d}{3}\times\frac{h}{4}$ **c** $\frac{2y}{8}\times\frac{2q}{3}$

d $\frac{x}{11}\times\frac{6y}{5}$ **e** $\frac{4k}{7}\times\frac{14v}{4}$ **f** $\frac{5h}{3}\times\frac{9d}{15}$

PATH

g $\frac{12}{v}\times\frac{2}{9}$ **h** $\frac{6}{5b}\times\frac{10b}{18}$ **i** $\frac{4}{5y}\times\frac{7y}{12}$

j $\frac{14}{15p}\times\frac{3hp}{7}$ **k** $\frac{20a}{3k}\times\frac{9a}{5k}$ **l** $\frac{5}{12b}\times\frac{8ab}{15}$

2 Simplify the product $\frac{6ad}{7m}\times\frac{3y}{4d}\times\frac{14gm}{27ay}$. Select the correct answer **A**, **B**, **C** or **D**.

A $\frac{2ak}{7y}$ **B** $\frac{g}{3}$ **C** $\frac{14gm}{9}$ **D** $\frac{7m}{4}$

3 Simplify the quotient $\frac{5xy}{3k}\div\frac{20}{9ky}$. Select **A**, **B**, **C** or **D**.

A $\frac{3y}{4x}$ **B** $\frac{4x}{15y^2}$ **C** $\frac{45k}{60y}$ **D** $\frac{3xy^2}{4}$

4 Simplify each quotient.

a $\frac{h}{5}\div\frac{4}{9}$ **b** $\frac{x}{4}\div\frac{3p}{7}$ **c** $\frac{k}{6}\div\frac{a}{11}$

d $\frac{8my}{27}\div\frac{4m}{9}$ **e** $\frac{3c}{4}\div\frac{5c}{8}$ **f** $\frac{2m}{9}\div\frac{4am}{3}$

PATH

g $\frac{10e}{3}\div\frac{4}{e}$ **h** $\frac{8}{3xy}\div\frac{6x}{5y}$ **i** $\frac{w}{y}\div\frac{4}{y}$

j $\frac{8k}{15gh}\div\frac{12hk}{10g}$ **k** $\frac{6d^2}{25}\div\frac{3d}{10n}$ **l** $\frac{8y}{5}\div\frac{32y^2}{3g}$

5 Simplify each expression.

a $\frac{wy}{5}\div\frac{3y}{20}$ **b** $\frac{d}{3h}\div\frac{k}{6}\div\frac{2d}{hk}$ **c** $\frac{24pq}{7}\div 8p$

d $\frac{x}{4y}\times\frac{12m}{5g}\times\frac{10y}{9x}$ **e** $\frac{2w}{p}\times\frac{3h}{7}\div\frac{4h}{5p}$ **f** $10pg\div\frac{6p}{5}$

g $\frac{4k}{3m}\div\frac{3d}{m}\div\frac{5k}{2m}$ **h** $\frac{3d}{4}\div\frac{d}{a}\times\frac{4d}{15}$ **i** $\frac{8}{3p}\div\frac{4}{a}\times\frac{p}{6a}$

j $15g\div\frac{3g}{4x}$ **k** $\frac{5}{2y}\times\frac{4}{3q}\div\frac{10}{3y}$ **l** $\frac{12}{b}\div\frac{4}{ab}$

6 Simplify the expression $\frac{4km}{5}\div\frac{18d}{7}\div\frac{m}{30}$. Select **A**, **B**, **C** or **D**.

A $\frac{175}{12km}$ **B** $\frac{7km^2}{1350d}$ **C** $\frac{28k}{3d}$ **D** $\frac{4km}{5d}$

Foundation Standard Complex

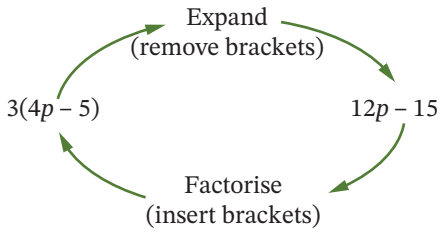
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Expanding and factorising expressions 4.04

Expanding and **factorising** are inverse operations.

When $3(4p - 5)$ is **expanded**, the answer is $12p - 15$.

When $12p - 15$ is **factorised**, the answer is $3(4p - 5)$



Expanding an expression

Multiply each term inside the brackets by the term outside.

$$a(b + c) = ab + ac$$

$$a(b - c) = ab - ac$$

Example 6

Expand each expression.

a $4d(5d + 3)$ **b** $-5(3p - 8)$

Solution

a $4d(5d + 3) = 4d \times 5d + 4d \times 3$ Multiplying out the brackets

$= 20d^2 + 12d$

b $-5(3p - 8) = -5 \times 3p - (-5) \times 8$ Multiplying out the brackets

$= -15p - (-40)$

$= -15p + 40$

Example 7

Expand and simplify by collecting **like terms**.

a $-3r^2(4r + 2) - 5r^3$ **b** $9(m - 3) + m(m - 10)$

Solution

a $-3r^2(4r + 2) - 5r^3 = -3r^2 \times 4r + (-3r^2) \times 2 - 5r^3$ Expanding

$= -12r^3 + (-6r^2) - 5r^3$

$= -17r^3 - 6r^2$ Collecting like terms to simplify

b $9(m - 3) + m(m - 10) = 9 \times m - 9 \times 3 + m \times m - m \times 10$

$= 9m - 27 + m^2 - 10m$

$= m^2 + 9m - 10m - 27$ It's neater to place m^2 at the front

$= m^2 - m - 27$ Collecting like terms to simplify

Video
Factorising expressions 2

Factorising an expression

- Find the **highest common factor (HCF)** of the terms and write it outside the brackets
- Divide each term by the HCF and write the answers inside the brackets
- $ab + ac = a(b + c)$ $ab - ac = a(b - c)$
- To check that the factorised answer is correct, expand it.

Example 8

Factorise each expression.

a $18xy^2 - 24xy$ **b** $3m(5 + 2d) + 7(5 + 2d)$ **c** $-5k^2 + 15k$

Solution

a The HCF of $18xy^2$ and $24xy$ is $6xy$.

$\therefore 18xy^2 - 24xy = 6xy \times 3y - 6xy \times 4$ Rewrite the expression using the HCF $6xy$

$= 6xy(3y - 4)$ Write the HCF at the front of the brackets

b The HCF of $3m(5 + 2d) + 7(5 + 2d)$ is $(5 + 2d)$.

$\therefore 3m(5 + 2d) + 7(5 + 2d) = (5 + 2d) \times 3m + (5 + 2d) \times 7$

$= (5 + 2d)(3m + 7)$

c When factorising expressions that begin with a negative term, we use the 'negative' HCF.

The highest 'negative' common factor of $-5k^2$ and $15k$ is $-5k$.

$\therefore -5k^2 + 15k = (-5k) \times k + (-5k) \times (-3)$ $(-5k) \times (-3) = +15k$

$= (-5k)[k + (-3)]$

$= -5k(k - 3)$

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PATH

4.04

Example 9

Factorise each expression.

a $8a^3 + 4a^2$ **b** $20h^3k + 25h^4k - 10h^2k$

Solution

a The HCF of $8a^3$ and $4a^2$ is $4a^2$.

$\therefore 8a^3 + 4a^2 = 4a^2 \times 2a + 4a^2 \times 1$ Rewrite the expression using the HCF $4a^2$

$= 4a^2(2a + 1)$

b The HCF is $5h^2k$.

$\therefore 20h^3k + 25h^4k - 10h^2k = 5h^2k \times 4h + 5h^2k \times 5h^2 - 5h^2k \times 2.$

$= 5h^2k(4h + 5h^2 - 2)$

EXERCISE 4.04 ANSWERS ON P. 586

Expanding and factorising expressions U F

1 Expand each expression. EXAMPLE 6

a $5(d + 11)$ **b** $-3(r + 10)$ **c** $7(x - 9y)$ **d** $-4(a - 5w)$

e $-(2 - p^2)$ **f** $-10e(2e^2 + 3)$ **g** $6y(1 + 7y)$ **h** $4xy(3xy - 1)$

i $8rq(2q - r)$ **j** $3ab(4b - 7a)$ **k** $-6h^2(1 - 3h)$ **l** $-5x(5x^2 + 4y)$

m $-(3 + 8a)$ **n** $-2m^2(3m - 4n)$ **o** $5g(3 + 7g^2)$ **p** $-1(5e - 12)$

2 Expand $-2y(5 + 7y)$. Select the correct answer **A**, **B**, **C** or **D**.

A $3y - 5y^2$ **B** $-10y + 14y^2$ **C** $-10y - 14y^2$ **D** $-10y - 5y^2$

3 Use the substitution $x = 2$ to test whether each equation is correct or incorrect.

a $4(x + 10) = 4x + 40$ **b** $5(x - 1) = 5x - 6$ **c** $x(3 - x) = 3x - x^2$

4 Expand and simplify by collecting like terms. EXAMPLE 7

a $4k(3k - 5) - 9k^2$ **b** $5h - 7h(4 - h)$

c $9w^3 - 3w(5 + 2w^2)$ **d** $24x^3 - 5x^2(2x^2 - 5x)$

e $8 - 3(2 - 7d)$ **f** $4n(3 - 5n) - 6n^2$

g $4y(y - 4) + 5(2y + 1)$ **h** $5(1 - 2a) - a(3 + 4a)$

i $7(3 + 6w) - (5 - 8w)$ **j** $4y^2(5y + 5) + 4(2 - 7y^2)$

k $-v(2v + 7) + 6(v - 1)$ **l** $2(4 - 3a) - a(3 - a)$

m $2c(5c - 1) - 4(7c - 5)$ **n** $3m(m + 5m^2) - m^2(1 - 3m)$

o $4x(2y + 5) - 6y(10 - 3x)$

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EXAMPLE 8

5 Factorise each expression.

a $15y - 20$ **b** $21 + 35w$

c $2p + p^2$ **d** $30y - 20y^2$

e $36d^2 + 24d$ **f** $28k^2 - 21k$

g $8(c - 5) - c(c - 5)$ **h** $m(3 + 2m) + 7(3 + 2m)$

i $-q^2 - 36q$ **j** $-8x + 12x^2$

k $b(3b + 5) - 2(3b + 5)$ **l** $-12cd^2 + 8cd$

m $-hn^2 + h^2n$ **n** $-15g^2 - 18g$

o $48q^2 - 54q$

6 Factorise $-16pw + 10xw$. Select **A**, **B**, **C** or **D**.

A $-2w(8p - 10x)$ **B** $-2w(8p - 5x)$

C $-8w(2p - x)$ **D** $-4w(5p + 2x)$

PATH

EXAMPLE 9

7 Factorise each expression.

a $8m^2y^2 - 12my$ **b** $36ab^2c + 27bc$

c $24m^2n - 108mn^2$ **d** $20dg^2 - 35ag$

e $40wy^3 + 24w^2y^2$ **f** $75g^3h^2 - 125gh$

g $-4p^3 - 8p^2 + p$ **h** $6mn^2 + 3mn + 48m^2n$

i $32p^3g + 8pg^2 - 8pg$ **j** $18a^5 - 12a^2 + 15a^4$

k $28m^3h^2 - 21mh^2$ **l** $15kwp - 24wp^2 - 9kw$



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Did you know?

CAPTCHA

For security purposes on a website, have you ever been asked to enter letters, words or numbers that have been displayed in a wavy, difficult-to-read format like this?



This process is called **CAPTCHA**, which stands for **Completely Automated Public Turing Test to Tell Computers and Humans Apart**, invented in 1999. It is designed to ensure that the person accessing the website is a human and not another computer that could be hacking into the site, sending spam or viruses. CAPTCHA uses optical character recognition (OCR), which is a technology that can convert images of text into editable text.

How does CAPTCHA prove that you are a human and not a computer?

What is a Turing test, named after English computer scientist and mathematician Alan Turing?



iStock.com/milindri

Quiz
Mental skills 4

Mental skills 4: Maths without calculators ANSWERS ON P. 587

Estimating answers

A quick way of estimating an answer is to round each number in the calculation.

1 Study each example.

a $55 + 132 - 34 + 17 - 78 \approx 60 + 130 - 30 + 20 - 80$
$= (60 + 20 - 80) + (130 - 30)$
$= 0 + 100$
$= 100$ (Actual answer = 92)

b $78 \times 7 \approx 80 \times 7$
$= 560$ (Actual answer = 546)

c $510 \div 24 \approx 500 \div 20$
$= 50 \div 2$
$= 25$ (Actual answer = 21.25)

2 Now estimate each answer.

a $27 + 11 + 87 + 142 + 64$
b $55 + 34 - 22 - 46 + 136$
c $684 + 903$
d $35 + 81 + 110 + 22 + 7$
e $517 - 96$
f $210 - 38 - 71 + 151 - 49$
g $766 - 353$
h 367×2
i 83×81
j 984×16
k $828 \div 3$
l $507 \div 7$

3 Study each example involving decimals.

a $20.91 - 11.3 + 2.5 \approx 21 - 11 + 3$
$= 13$ (Exact answer = 12.11)

b $4.78 \times 19.2 \approx 5 \times 20$
$= 100$ (Exact answer = 91.776)

c $\frac{37.6+9.3}{41.2-12.7} \approx \frac{38+9}{40-13}$
$= \frac{47}{27}$
$= \frac{50}{30}$
$= 1.6$ (Exact answer = 1.6456...)

4 Now estimate each answer.

a $3.75 + 9.381 + 4.6 + 10.5$
b $14.807 + 6.6 - 7.22$
c 18.47×9.61
d 4.27×97.6
e $\frac{11.07+18.4}{12.2}$
f $\frac{38.18}{17.2-9.6}$
g $\frac{18.46 \times 4.9}{39.72-15.2}$
h $62.13 \div 10.7$
i $(4.89)^2$

Expanding binomial products

4.05

Binomial means '2 terms'. $(m + 8)$ and $(m - 3)$ are **binomial expressions** because they each have exactly 2 terms. $(m + 8)(m - 3)$ is called a **binomial product** because it is a product of 2 binomial expressions.

Example 10

Expand each binomial product.

a $(m + 8)(m - 3)$

b $(4y - 5)(3y - 2)$

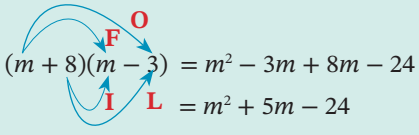
Solution

a $(m + 8)(m - 3) = m(m - 3) + 8(m - 3)$ Each term in $(m + 8)$ is multiplied by $(m - 3)$

$= m^2 - 3m + 8m - 24$ Expanding

$= m^2 + 5m - 24$ Simplifying

One way of remembering which pairs of terms to multiply together in a binomial product is called the **FOIL method**, as shown.



- **F** means multiply the **f**irst terms: $m \times m = m^2$
- **O** means multiply the **o**utside terms: $m \times (-3) = -3m$
- **I** means multiply the **i**nside terms: $8 \times m = 8m$
- **L** means multiply the **l**ast terms: $8 \times (-3) = -24$

b $(4y - 5)(3y - 2) = 12y^2 - 8y - 15y + 10$ Using FOIL

$= 12y^2 - 23y + 10$ Simplifying

Videos
Expanding binomial products 1
Expanding binomial products 2

Worksheets
Mixed expansions
Algebra 2
Binomial products
Area diagrams

Puzzles
Expanding brackets
Trinominoes

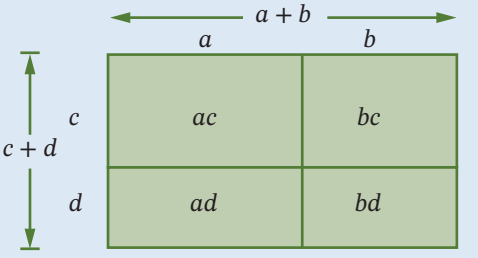
Technology
Expanding binomials

Spreadsheet
Expanding binomials

Expanding a binomial product

Multiply each term in the first binomial by each term in the second binomial.

$$(a + b)(c + d) = ac + ad + bc + bd$$



EXERCISE 4.05 ANSWERS ON P. 587

Expanding binomial products U F R C

EXAMPLE 10

1 Expand each binomial product.

a	$(m + 4)(m + 3)$	**b**	$(w + 5)(w + 5)$	**c**	$(y + 12)(y - 12)$
d	$(h + 7)(h - 9)$	**e**	$(a - 5)(a + 3)$	**f**	$(x - 11)(x - 4)$
g	$(p + 3)(p + 8)$	**h**	$(c - 7)(c - 12)$	**i**	$(g - 1)(g - 2)$
j	$(u - 8)(u + 7)$	**k**	$(m + 4)(m - 10)$	**l**	$(q - 11)(q + 6)$
m	$(d - 5)(8 + d)$	**n**	$(10 - e)(e - 7)$	**o**	$(9 - h)(5 - h)$

2 Expand $(w - 8)^2$. Select the correct answer **A**, **B**, **C** or **D**.

A $w^2 - 64$ **B** $w^2 + 64w$ **C** $w^2 + 16w - 64$ **D** $w^2 - 16w + 64$

3 Expand $(12 - a)^2$. Select **A**, **B**, **C** or **D**.

A $144 - a^2$ **B** $144 - 24a + a^2$

C $a^2 + 24a + 144$ **D** $a^2 + 144$

4 Expand each binomial product.

a	$(3y + 1)(y + 7)$	**b**	$(4k + 9)(3k + 2)$	**c**	$(3m - 5)(m + 3)$
d	$(5p + 3)(2p - 5)$	**e**	$(2w - 3)(w - 8)$	**f**	$(7x + 4)(2x + 3)$
g	$(3b - 4)(3b - 4)$	**h**	$(5a + 6)(5a + 6)$	**i**	$(2q - 7)(5q + 7)$
j	$(6 + 5p)(p - 3)$	**k**	$(4d - 11)(1 + 3d)$	**l**	$(2r - 5)(9 - 4r)$
m	$(7y + 3)(3 - 2y)$	**n**	$(8h - 3)(8h + 3)$	**o**	$(9 - 7w)(7w - 9)$
p	$(4d - 1)(4d + 1)$	**q**	$(f - 1)(1 + 3f)$	**r**	$(6u - 5)(5 - 6u)$

5 Expand $(5 - 3k)(5 + 3k)$. Select **A**, **B**, **C** or **D**.

A $25 - 30k - 9k^2$ **B** $25 - 9k^2$

C $9k^2 - 30k - 25$ **D** $9k^2 - 25$

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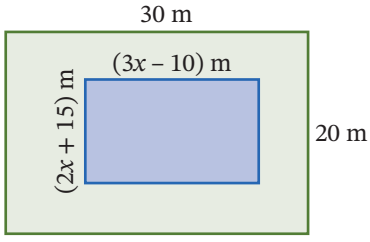
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6 Expand each binomial product and notice any patterns. R C

a	$(x+7)^2$	**b**	$(x-7)^2$	**c**	$(x+7)(x-7)$
d	$(y-6)(y+6)$	**e**	$(y-6)^2$	**f**	$(y+6)^2$
g	$(11+y)(11-y)$	**h**	$(11+y)^2$	**i**	$(11-y)^2$
j	$(3b+7)^2$	**k**	$(3b+7)(3b-7)$	**l**	$(3b-7)^2$

Chapter 12, *Products and factors*, explores these special products more.

7 This diagram shows a house $(3x-10)$ m long and $(2x+15)$ wide on a block of land with dimensions 30 m × 20 m. R C

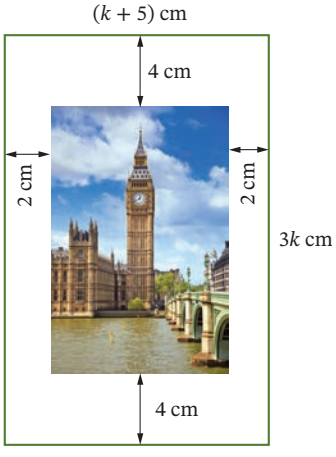


a Write down a binomial expression for the area of the house in square metres.

b Expand and simplify your expression for the area.

c The green area of the block of land not covered by the house is to be turfed. Write a simplified expression for this area in square metres.

8 A photograph frame is $(k+5)$ cm wide and $3k$ cm long. The gap between the photograph and frame is 4 cm at the top and bottom and 2 cm on each side. R C



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a What is the area of the photograph frame?

b Write down expressions for the length and width of the photograph.

c Write down a binomial expression for the area of the photograph.

d Expand and simplify your expression for the area of the photograph.

e Find an expression for the area of the frame not taken up by the photograph.

9 A family room in a house is to be extended. The room is a metres long and b metres wide. The length is to be increased by 3 metres and the width by 1 metre. R C

a Write down expressions for the new length and width in metres.

b Write down a binomial expression for the new area of the room in square metres.

c Expand and simplify your expression for the area.

d By how much has the area of the room increased in square metres?

10 Prove that: R C

a $(a-b)^2 = (b-a)^2$

b $(a+b)(a-b) = a^2 - b^2$

c $(a-b)^2 = a^2 - 2ab + b^2$

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Investigation

Factorising quadratic expressions

1 **a** Show that $(x + 3)(x + 5) = x^2 + 8x + 15$.

b The quadratic expression $x^2 + 8x + 15$ has 3 terms. The coefficient of x (the number in front of the x) is 8. How are the 3 and 5 in $(x + 3)(x + 5)$ related to the 8?

c The constant term in $x^2 + 8x + 15$ is 15, the number with no x at the end. How are the 3 and 5 related to the 15?

2 **a** Expand $(x + 9)(x + 2)$.

b What is the coefficient of x? How are 9 and 2 related to it?

c What is the constant term? How are 9 and 2 related to it?

3 **a** Expand $(x + 8)(x - 3)$.

b What is the coefficient of x? How are 8 and -3 related to it?

c What is the negative constant term? How are 8 and -3 related to it?

4 **a** Expand $(x - 4)(x - 1)$.

b What is the negative coefficient of x? How are -4 and -1 related to it?

c What is the positive constant term? How are -4 and -1 related to it?

5 In the expansion of any binomial product, how are the coefficient of x and the constant term related to the numbers in the binomials?

6 Copy and complete:

a $(x + ___)(x + ___) = x^2 + 5x + 4$

b $(x + ___)(x + ___) = x^2 + 8x + 15$

c $(x + ___)(x + ___) = x^2 + 7x + 12$

d $(x + ___)(x - ___) = x^2 - 4x - 32$

e $(x + ___)(x - ___) = x^2 + 2x - 3$

f $(x - ___)(x - ___) = x^2 - 9x + 20$

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Factorising quadratic expressions $x^2 + bx + c$

4.06

PATH

4.06

A **quadratic expression** is an algebraic expression in which the highest power of the variable is 2, such as $x^2 - 5x + 7$, $x^2 - 15$, $2x^2 - 3x + 9$ and $-4x^2 + 7x$.

A quadratic expression such as $x^2 - 5x + 7$ is called a **trinomial** because it has 3 terms.

The expansion of $(x + 2)(x + 4)$ is $x^2 + 6x + 8$, a quadratic trinomial.

$\therefore$ The factorisation of $x^2 + 6x + 8$ is $(x + 2)(x + 4)$.

Factorising quadratic trinomials

In the factorisation of a quadratic trinomial such as $x^2 + 6x + 8$:

- each factor must have an x term to give x^2

$x^2 + 6x + 8 = (x + 2)(x + 4)$

- $2 + 4 = 6$, which is the **coefficient** of x, the number in front of the x

$x^2 + 6x + 8 = (x + 2)(x + 4)$

- $2 \times 4 = 8$, which is the **constant term** with no x

$x^2 + 6x + 8 = (x + 2)(x + 4)$

Example 11

Factorise each quadratic trinomial.

a $a^2 + 7a + 12$ **b** $x^2 + 9x + 8$

Solution

a Find the 2 numbers that have a sum of 7 and a product of 12.

It is best to test numbers that have a **product of 12** and then check if their sums equal 7.

Pair of numbers	Product	Sum
6, 2	$6 \times 2 = 12$	$6 + 2 = 8$
3, 4	$3 \times 4 = 12$	$3 + 4 = 7$

The correct numbers are 3 and 4.

$\therefore a^2 + 7a + 12 = (a + 3)(a + 4)$

b Find 2 numbers with a sum of 9 and a product of 8.

Test numbers that have a **product of 8** and check if their sums equal 9.

Pair of numbers	Product	Sum
2, 4	$2 \times 4 = 8$	$4 + 2 = 6$
8, 1	$8 \times 1 = 8$	$1 + 8 = 9$

The correct numbers are 8 and 1.

$\therefore x^2 + 9x + 8 = (x + 8)(x + 1)$

Video
Factorising quadratic expressions 1

PATH

Factorising quadratic expressions of the form $x^2 + bx + c$

- Find 2 numbers that have a sum of b and a product of c
- Use these 2 numbers to write a binomial product of the form $(x ___)(x ___)$

Video
Factorising quadratic expressions 1

Example 12

Factorise each quadratic expression.

a $x^2 + x - 6$ **b** $a^2 - 2a - 15$ **c** $y^2 - 6y + 8$

Solution

a $x^2 + x - 6$
Find 2 numbers that have a product of -6 and a sum of 1.
Since the product is negative, one of the numbers must be negative.
They are $+3$ and -2.
$\therefore x^2 + x - 6 = (x + 3)(x - 2)$

b $a^2 - 2a - 15$
Product $= -15$, sum $= -2$.
Since the product is negative, one of the numbers must be negative.
They are -5 and $+3$.
$\therefore a^2 - 2a - 15 = (a - 5)(a + 3)$

c $y^2 - 6y + 8$
Product $= 8$, sum $= -6$.
Since the sum is negative, one of the numbers must be negative.
Since the product is positive, *both* of the numbers must be negative.
They are -2 and -4.
$\therefore y^2 - 6y + 8 = (y - 2)(y - 4)$

EXERCISE 4.06 ANSWERS ON P. 587

Factorising quadratic expressions $x^2 + bx + c$ U F R

EXAMPLE 11

1 Find 2 numbers whose: R

a product is -15 and sum is 2
b product is 40 and sum is -14
c product is 42 and sum is 13
d product is -6 and sum is -1
e product is 45 and sum is 14
f product is -24 and sum is 2
g product is 54 and sum is 15
h product is -16 and sum is 0
i product is -10 and sum is 3
j product is 35 and sum is -12

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EXAMPLE 12

4.06

2 Factorise each quadratic expression. R

a $y^2 + 8y + 12$ **b** $m^2 + 15m + 56$ **c** $g^2 + 9g + 14$

d $w^2 + 13x + 36$ **e** $p^2 + 11p + 24$ **f** $a^2 + 13a + 42$

g $e^2 + 12e + 27$ **h** $n^2 + 6n + 9$ **i** $c^2 + 10c + 21$

3 Factorise each quadratic trinomial. R

a $x^2 - 9x + 20$ **b** $h^2 - 13h + 30$ **c** $p^2 - 11p + 24$

d $e^2 - 11e + 30$ **e** $w^2 - 17w + 72$ **f** $k^2 - 10k + 9$

g $m^2 - 16m + 64$ **h** $u^2 - 5u + 6$ **i** $d^2 - 12d + 35$

4 Factorise each quadratic expression. R

a $q^2 - 8q - 20$ **b** $h^2 - 5h - 36$ **c** $y^2 + 7y - 44$

d $x^2 - 2x - 63$ **e** $u^2 + 9u - 10$ **f** $e^2 + 7e - 30$

g $a^2 - a - 110$ **h** $y^2 + 6y - 27$ **i** $m^2 - 6m - 7$

j $c^2 + 7c - 18$ **k** $k^2 + 3k - 54$ **l** $r^2 - 9r - 22$

m $p^2 - 4p - 32$ **n** $u^2 + 12u - 45$ **o** $b^2 - 6b - 16$

5 Factorise $a^2 - 11a - 42$. Select the correct answer **A**, **B**, **C** or **D**.

A $(a - 7)(a + 6)$ **B** $(a + 7)(a - 8)$ **C** $(a - 21)(a + 2)$ **D** $(a + 3)(a - 14)$

6 Factorise each quadratic trinomial. R

a $h^2 - 2h + 1$ **b** $x^2 + 15x + 50$ **c** $r^2 + 20r + 96$

d $a^2 - 3a - 28$ **e** $u^2 - 7u - 60$ **f** $y^2 - 18y + 81$

g $v^2 - v - 56$ **h** $w^2 - 11w - 60$ **i** $g^2 + 3g - 18$

j $p^2 + 14p + 48$ **k** $e^2 + 7e - 8$ **l** $x^2 - 19x + 84$

Power plus ANSWERS ON P. 587

1 Simplify each expression using index notation.

a $13^7 \times (13^5 \div 13^{11})$ **b** $(8^{-4})^5 \div (8^5)^{-4}$ **c** $\left(\frac{(7^2)^3}{7^5 \times 7^2}\right)^{-1}$

d $\left(\frac{1}{10}\right)^{-10} \times \left(\frac{1}{10}\right)^{10}$ **e** $\left(-2\frac{1}{2}\right)^3$ **f** $\left(\frac{1}{2}\right)^{-10} - \left(\frac{1}{4}\right)^{-5}$

2 Simplify each expression.

a $\frac{1}{x} + \frac{1}{y} - \frac{1}{w}$ **b** $\frac{1}{x} \times \frac{1}{y} \div \frac{1}{w}$ **c** $\frac{1}{x} \times \left(\frac{1}{y} + \frac{1}{w}\right)$

3 Expand each expression.

a $(a - b)(a + b - c)$ **b** $(x - y + 1)^2$ **c** $t^{-1}(3t^2 + 4t - 1)$

4 Factorise each expression.

a $x^2 - 133x + 1000$ **b** $y^2 + 14y - 1800$

c $b^2 + 82b + 1681$ **d** $n^2 - 2500$

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CHAPTER 4 REVIEW

Quiz
Language of maths 4

Puzzle
Algebra crossword

Language of maths

algebraic fraction	base	binomial	brackets
coefficient	constant term	denominator	expand
expression	factorise	highest common factor	index law
indices	numerator	product	power
quadratic expression	reciprocal	term	variable

1 Dividing by a fraction is the same as multiplying by its what?

2 Any number raised to the power of 0 is equal to what?

3 What is the difference between **expand** and **factorise?**

4 What power is associated with the reciprocal of a term or number?

5 In the quadratic trinomial $2x^2 - 3x + 6$, what is:

a the constant term? **b** the coefficient of x?

6 Copy and complete:

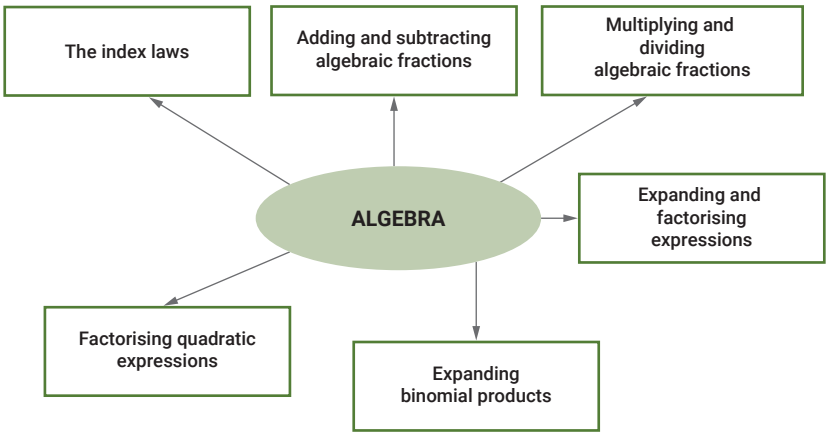
To factorise quadratic expressions of the form $x^2 + bx + c$, first find 2 numbers that have a _____ of b and a _____ of c.

Worksheet
Mind map: Algebra

Topic summary

- What was this topic about? What was the main theme?
- What content was new and what was revision for you?
- Write 3 index laws in both words and symbols.
- Write 10 questions (with solutions) that could be used in a test for this chapter.
  Include some questions that you have found difficult to answer.
- List the sections of work in this chapter that you did not understand. Follow up this work.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



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TEST YOURSELF 4

ANSWERS ON P. 588

Quiz
Test yourself 4

4.01

PATH

TEST YOURSELF 4

1 Simplify each expression.

a $3v^4 w^2 \times 2vw^5$

b $\dfrac{24t^8h^8}{3th^2}$

c $(5xy^2)^2$

d $\left(\dfrac{2p}{3}\right)^0$

e $(4k)^{-1}$

f $\left(\dfrac{5y}{2}\right)^3$

g $8^0 - 2^0$

h $3^{-1} + 4^0$

i $16a^3g^5 \div 18ag^4$

4.01

2 Simplify each expression.

a $(4m)^{-2}$

b $4m^{-2}$

c $\left(\dfrac{5b^8y^6}{b^2y^3}\right)^4$

d $(4t^4 u^5)^3 \times 8t^2 u$

e $45c^6 d^8 \div (-3cd^2)^2$

f $\left(\dfrac{45ab^4}{54a^2b^3}\right)^{-1}$

4.02

3 Simplify each expression.

a $\dfrac{t}{4} - \dfrac{2t}{5}$

b $\dfrac{5g}{3} + \dfrac{3g}{2}$

c $\dfrac{5x}{16} - \dfrac{x}{4}$

d $\dfrac{3b}{7} + \dfrac{2}{7}$

e $\dfrac{5w}{8} - \dfrac{3}{2}$

f $\dfrac{7}{12} - \dfrac{y}{5}$

g $\dfrac{5m}{7} + \dfrac{3n}{4}$

h $\dfrac{5p}{6} - \dfrac{p}{8}$

4.03

4 Simplify each expression.

a $\dfrac{3}{m} \times \dfrac{p}{4}$

b $\dfrac{1}{q} \times \dfrac{3}{q}$

c $\dfrac{2d}{3} \times \dfrac{15y}{14d}$

d $\dfrac{m}{3} \div \dfrac{2m}{9}$

e $\dfrac{5}{2a} \div \dfrac{4a}{7}$

f $\dfrac{12md}{5} \div \dfrac{3m}{10d}$

g $\dfrac{4xy}{5q} \div \dfrac{3x}{2y} \times \dfrac{9q}{4y}$

h $\dfrac{8ab}{3c} \div \dfrac{5}{2c} \div \dfrac{4b}{15c}$

4.04

5 Expand each expression.

a $9(m - 8)$

b $b(10a + b)$

c $-3y(4x^2 - 5y)$

d $8tp(7p - 5t)$

e $-(3n - 10)$

f $-5h^2(3h + 7)$

g $4y(5y - 7h)$

h $-wx\,(3x - 7w)$

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4.04

6 Expand and simplify each expression.

a $5g - 3g(6 - 7g)$

b $4fg(g - 6f) - 6f^2 g$

c $12(9 - n) - 5(2n + 3)$

d $x^2(6x + x^2) + 2x(3x^3 + x^2)$

e $3(7 - 2y) - 5y(7 - 2y)$

4.04

7 Factorise each expression.

a $8t - 72$

b $b^2 + 36b$

c $-3m - 33$

d $36wr^2 + 28w^2r$

e $-24p + 18q$

f $2(5x - 1) - 3x(5x - 1)$

PATH

4.04

8 Factorise each expression.

a $15xy^2 - 30x^3 y^3$

b $6pt^2 + 12p^2 t - 48p^3$

c $32r^2 s^4 + 12r^4 s^3$

d $50x^4 y^3 - 75x^3 y^4$

e $-8p^3 q^3 + 48p^3 q^6$

f $n(n^2 + 6) - (n^2 + 6)$

4.05

9 Expand each binomial product.

a $(b + 3)(b + 10)$

b $(d + 8)(d - 7)$

c $(t - 6)(9 - t)$

d $(5x + 7)(4x - 3)$

e $(7y - 5)(7y + 5)$

f $(3p - 8)(7p - 2)$

g $(3m + 7)(3m + 7)$

h $(4 - x)(3x + 2)$

i $(6 - 5d)(3 - 2d)$

PATH

4.06

10 Factorise each quadratic expression.

a $y^2 + 10y + 25$

b $x^2 - 21x + 20$

c $n^2 + 8n - 33$

d $a^2 - 11a + 28$

e $m^2 - 5m - 84$

f $p^2 + 3p - 54$

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